Progressive Learning-Based Jitter Distortion Correction for Remote Sensing Images of Time Delay and Integration Camera

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***Abstract*— The widespread use of time delay and integration charge-coupled device (TDI CCD) technology in high-resolution spaceborne optical cameras has made high-frequency jitter effects a common issue, resulting in different levels of distortion in images. Current methods mostly concentrate on the correction of obviously high levels of geometric distortion. Focusing on low levels of geometric distortion, which are more difficult to detect accurately, this article proposes a progressive learning-based correction method for high-frequency jitter distortion in remote sensing images from spaceborne TDI CCD cameras using a generative adversarial network (GAN). First, a distorted dataset with diverse jitter levels for progressive training is generated through the jitter simulation model by adjusting the parameters. Then, a GAN model is employed for the correction task. The generator consists of the distortion net for geometric distortion correction and the detail enhancement net for image detail restoration. Finally, a progressive learning strategy is used to gradually enhance the ability of the network to correct minor geometric distortion. The proposed method is validated using simulated images and real-world satellite images. Experimental results demonstrate that the proposed method outperforms existing restoration methods both in simulated datasets and practical scenarios.**

***Index Terms*— Distortion correction, generative adversarial network (GAN), image restoration, satellite jitter, time delay and integration charge-coupled device (TDI CCD).**

I. Introduction

THE advancement of remote sensing technology has significantly improved the resolution of optical sensors, leading to the widespread use of high-resolution satellites in fields like mapping and monitoring [1]. Satellite platform jitter refers to the disturbances caused by factors such as attitude adjustment, pointing control, solar panel adjustment, and the periodic movement of components on the satellite during its orbital operation. These disturbances cause a small-amplitude jitter of the satellite [2]. Although subtle and time-varying, such high-dynamic disturbances still exert a considerable impact on the quality of high-resolution remote sensing imagery. Many satellites experience platform jitter issues, such as advanced land observing satellite (ALOS) [3], Pleiades [4], ZY-3 satellite [5], [6], and GF-1 02/03/04 satellites [7], [8].

With the rapid development of optical remote sensing satellites, the demand for higher spatial resolution is growing. In order to meet the demand for high resolution, low cost, lightweight design, and time delay and integration charge-coupled devices (TDI CCDs) have become a critical photosensitive device in high-resolution satellites. TDI CCD is a special type of linear CCD that use multistage time integration to enhance sensitivity and improve the signal-to-noise ratio under low-light conditions [9]. The use of integral imaging techniques, however, makes TDI CCDs more susceptible to satellite jitter [10], [11]. The sensitivity and randomness of satellite jitter make it challenging to eliminate its effects through physical methods. When a satellite jitter is transmitted to the image plane, it induces offsets that degrade the geometric and radiometric properties of the image, resulting in a deterioration in image quality. Additionally, TDI CCDs are particularly prone to jitter-induced displacements during imaging, which exacerbates the mismatch between the speed of image motion and the rate of charge transfer, further degrading image quality [12]. As the resolution of satellite payloads continues to increase, the impact of platform jitter on the geometric accuracy of high-resolution satellite remote sensing images is becoming more pronounced.

Platform vibration is the primary factor affecting the image quality of TDI cameras. It induces image shifts in both the along-track and across-track directions, leading to geometric degradation and variations in the point spread function (PSF), which ultimately results in geometric distortion. Additionally, phase shifts caused by variations in the drift angle and mismatched charge transfer times further degrade the image quality.

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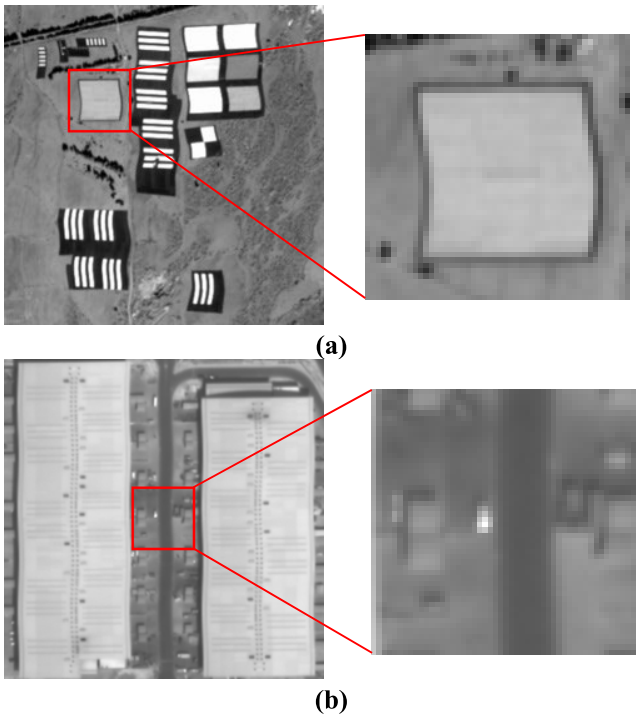


Fig. 1. Real remote sensing images with different level of jitter distortion. (a) High level of geometric distortion. (b) Lower level of geometric distortion.

The jitter characteristics encountered by different satellites vary, leading to differences in the extent of geometric distortion they experience. As shown in Fig. 1(a), this is a real distorted remote sensing image that exhibits a high level of geometric distortion but relatively low level of blur. As shown in Fig. 1(b), this is a really distorted remote sensing image that exhibits a lower level of geometric distortion but a higher level of blur.

The image degradation process is inherently complex, and the extent of degradation varies across each distorted image. Previous methods were primarily based on inversely estimating degradation parameters based on a rigorous degradation model for correction [13]. This approach, however, requires high accuracy in parameter inversion, as the correction precision is highly sensitive to the model parameters. Traditional methods, such as image resampling and estimating the PSF for the entire image, are unsuitable due to the complexity of the degradation process [14], [15]. The degradation factors are coupled, making it difficult to estimate the PSF. In most practical cases, only a single degraded image is available, and the exact degradation parameters are unknown, limiting the effectiveness of these traditional methods. Traditional deconvolution techniques, moreover, may be unsuitable, as they cannot handle the distortions resulting from the coupled degradation factors.

In order to address this issue, we propose a method for correcting geometric distortions in spaceborne TDI CCD camera images caused by platform jitter based on a generative adversarial network (GAN). The main innovation of this article lies in the following.

1) By analyzing the imaging theory of platform jitter for spaceborne TDI CCD cameras, we propose a jitter simulation model that simulates diverse levels of distortion according to jitter levels. The classification of jitter levels relies on amplitude and attenuation factors. Based on this model, a realistic large-scale image dataset with diverse levels of distortion was created for training.
2) In order to correct geometric distortions with diverse levels, a GAN model with the generator consisting of a distortion net and a detail enhancement net was employed. The Distortion Net primarily corrects geometric distortions and the detail enhancement net focuses on restoring image details.
3) In order to accurately detect and correct low levels of geometric distortions caused by high-frequency jitter, we proposed a progressive learning strategy to enhance the ability of the GAN model to detect and correct the minor distortions.

This article is organized as follows. In Section II, we provide a comprehensive overview of related work on platform jitter compensation and the application of deep learning in image restoration. We provide a detailed description of our proposed methodology in Section III. Section IV presents the results and experiments, including dataset generation, comparison results with other methods using both simulated and real distorted remote sensing images, and ablation studies. It also provides a detailed discussion of the experimental findings. Finally, Section V summarizes our research findings and discusses our future work.

II. Related Works

The quality of high-resolution remote sensing satellites is significantly affected by platform jitter. Many researchers have developed and implemented measures for jitter isolation and suppression during the design and manufacturing process of satellites [16], [17]. It is, however, impossible to completely eliminate jitter through physical means. Postprocessing remains necessary. Currently, the methods for correcting jitter errors can be generally categorized into three approaches: attitude sensor-based jitter measurement and correction; parallax image-based platform jitter detection, modeling, and compensation; and deep learning-based image restoration.

A. Attitude Sensor-Based Jitter Measurement and Correction

Attitude sensor-based jitter measurement methods rely on high-precision onboard angle sensors to capture platform motion parameters such as angular displacement, angular velocity, and angular acceleration. Wang et al. [18] proposed a jitter distortion correction method based on virtual steady reimaging, demonstrating its effectiveness in improving image accuracy. Xu et al. [19] devised a steady-state imaging model based on high-frequency angular displacement for jitter detection and compensation, achieving an internal geometric accuracy of 1.5 pixels after compensation. Zhao et al. [20] proposed a method for detecting jitter using a linear accelerometer and compared it with a parallax imaging-based model, validating its feasibility for on-orbit jitter detection.

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B. Parallax Image-Based Platform Jitter Detection, Modeling, and Compensation

Image-based jitter detection primarily relies on the imaging characteristics of optical satellites or reference data to identify jitter rules. Tong et al. [21] initially proposed a framework for jitter detection and compensation using images and identified a jitter frequency of approximately 0.67 Hz in the ZY-3 satellite using the comprehensive image matching approach [22]. Additionally, they introduced a subpixel phase correlation image matching method, which detected a jitter frequency of 0.65 Hz in the ZY-3 satellite using multispectral images [5]. Zhu et al. [6] detected platform jitter of 0.6 Hz in the ZY-3 satellite through least-squares dense matching using multispectral imagery; further research found that the GF-1 02/03/04 satellites suffered jitter in the range of 1.1–1.2 Hz in the cross-track direction [7], [8]. In addition, rolling shutter CMOS sensors have also been employed for platform jitter detection. Zhao et al. [23] analyzed and compensated for images from a high-resolution satellite using CMOS imaging techniques, improving the geometric accuracy of the images. Zhu et al. [24] proposed a method for detecting satellite platform jitter using sequence CMOS images with a rolling shutter, achieving 0.02-pixel detection accuracy.

According to Fourier transform theory, platform jitter can be modeled as a combination of sinusoidal functions with different frequencies, amplitudes, and phases [25]. Sun et al. [26] employed a jitter detection method based on short-time asynchronous satellite images to detect three different jitter frequencies of 0.1, 0.635, and 4 Hz in the along-track direction of Mapping Satellite-1. Pan et al. [27] deduced a jitter displacement estimation model based on the relationship between the jitter displacement and relative registration error, which was validated through simulations and real image experiments. The results show that the model achieves high accuracy. Research on jitter modeling for multistage TDI CCD images remains in its early stages. Zhu [28] established a jitter detection model for multistage integrated images and conducted preliminary verification through numerical and image simulation experiments. Ye et al. [1] proposed a method for detecting time-variant satellite jitter using multispectral images to obtain a parallax map, demonstrating high accuracy through experiments with ZY-3 satellite images. Pan et al. [29] further developed a continuous dynamic shooting model (CDSM) that reflects the real push-broom mode of TDI CCD, significantly improving the accuracy of jitter detection.

The compensation of jitter images depends on precise detection and modeling results, with degraded images being resampled to correct the parallax caused by platform jitter. Sun et al. [26] proposed a spatially varying blur model (SVBM) and view angle correction method (VACM) to compensate for the geometric and radiometric degradation of images. Pan et al. [29] proposed a context-adaptive recovery method based on the CDSM model, suitable for a wide range of jitter frequencies, significantly improving the geometric and radiometric quality of images. Liu et al. [30] proposed a jitter amplitude estimation model for TDI CCD images with attenuation compensation, providing a valuable reference for satellite jitter detection and estimation based on images.

C. Deep Learning-Based Image Restoration

Recent advancements in convolutional neural networks (CNNs) have significantly impacted the field of image restoration. These methods can generally be classified into three main types: CNN-based methods [31], transformer-based methods [32], [33], and GAN-based methods [34], [36].

Cho et al. [31] proposed MIMO-UNet, a novel single-image deblurring network that uses a single encoder for multiscale inputs and a single decoder for multioutput deblurring. Zamir et al. [32] introduced the restormer, an optimized transformer model for high-resolution image restoration. By employing multihead attention and feed-forward network, restormer achieves superior performance in tasks such as deraining, deblurring, and denoising. Wang et al. [33] proposed Uformer, a transformer-based model for image restoration. It incorporates a locally enhanced window transformer block for reduced complexity and a multiscale modulator for improved detail recovery. DeblurGAN [34] applied GAN to the task of image deblurring, treating deblurring as an image-to-image translation task [35]. DeblurGAN-v2 [36] built upon this by incorporating a feature pyramid network (FPN) and multiscale feature fusion in the generator, along with global (image) and local (patch) scales in the discriminator, further enhancing deblurring performance. Zhu et al. [37] developed a feature alignment-based method for single-image deblurring in remote sensing, improving performance through feature alignment fusion and feature importance selection modules.

Most image restoration tasks, however, concentrate on deblurring, and few focus on correcting geometric distortions. Wang et al. [38] introduced RestoreGAN, a network architecture that effectively learns and corrects deformation features caused by posture jitter, outperforming other methods in geometric distortion correction. Li et al. [39] presented a TDI CCD camera imaging simulation and learning-based image restoration approach, employing a neural network with a row-attention block and row-encoder block to achieve effective deblurring and geometric distortion correction, validated by real distorted images from Yaogan-26 satellite. Recently, Chen et al. [40] and Zhang et al. [41] developed a jitter-aware restoration network (JARNet), a two-stage restoration network that combines optical flow correction with spatial-frequency residual learning to mitigate geometric distortions and motion blur. They also develop a data synthesis pipeline to generate realistic jitter-degraded datasets, facilitating effective training of the network. Experimental results demonstrate that their method achieves effective correction on both simulated datasets and real remote sensing images.

The adverse effects of geometric distortion on image quality are, moreover, not limited to the field of remote sensing. When photographic equipment is used to capture underwater scenes through the turbulent water, surface disturbances also cause geometric distortion in the captured images. Li et al. [42] proposed a deep-learning approach to correct distortions in

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single images captured through turbulent water, using a warping and color prediction network to enhance image quality without the need for additional frames or reference images. This method provides a reference for correcting geometric distortions in remote sensing images caused by platform jitter in spaceborne TDI CCD cameras. Distortions from turbulent water are nonlinear and unpredictable, whereas distortions from TDI CCD cameras exhibit a more structured, line-by-line pattern. The causes of these two types of geometric distortions are different. Geometric distortions in underwater images result from disturbances on the water surface, while those in remote sensing images are caused by platform vibrations. In both cases, these distortions degrade image quality. This suggests that similar methods could be applied to address geometric distortion issues in remote sensing imagery.

Above all, image-based platform jitter detection methods depend on matching accuracy and suffer from frequency detection, which prevents them from fully covering the entire frequency spectrum of the jitter signal. Additionally, these methods require high computational complexity. In contrast, attitude sensor-based methods rely on high-performance sensors, leading to high costs. Accurate detection and modeling of jitter errors are crucial for effective compensation. The deviations in jitter model parameters may affect the compensation results. Traditional methods are, furthermore, highly dependent on detection accuracy and sensor sensitivity, which limits their practicality in many applications.

In addition, while researchers have made significant advances in deep learning-based image restoration techniques, most of these efforts focus on correcting motion blur in consumer cameras rather than addressing the geometric distortions caused by satellite platform vibrations. Although some studies have explored geometric distortion correction in remote sensing images, it remains uncertain whether these methods can effectively correct the subtle distortions caused by high-frequency jitter, which is currently more common.

III. Methodology

A. Jitter Error Imaging Simulation and Dataset Preparation

The aim of the imaging simulation was to study the impact of jitter on image quality by setting specific parameters. This approach ensured that the simulation results closely resembled real distorted images, thus ensuring the effectiveness of the training. As far as we are concerned, there is no specialized dataset available for jitter distortion correction; therefore, we selected a large number of images with distinct geometric features from various object detection datasets to serve as reference images for constructing our dataset. Fig. 2 illustrates the generation process of datasets with diverse levels of distortion. During the simulation process, we carefully considered the prominence of the geometric features present in the images to ensure that the simulated distorted images can provide clear distortion information that the network could capture.

The jitter curve can be modeled using the sine function, i.e., $\varepsilon(t) = A\sin(2\pi t/T + \phi)$. A and ϕ represent the amplitude and initial phase of image motion caused by platform jitter

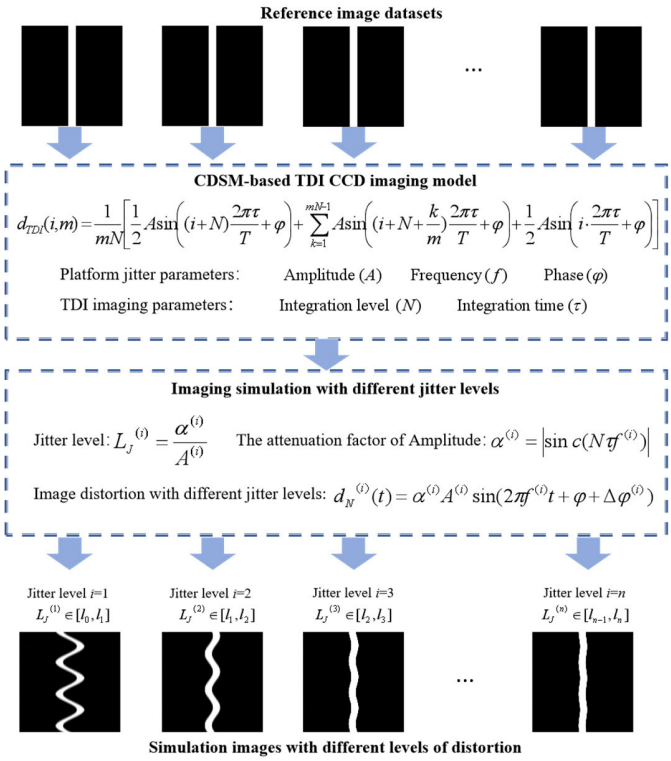


Fig. 2. Generation of datasets with different distortion levels.

under a single integration time, respectively. T is the period of the jitter curve. The final output image of the TDI CCD is a composite of multistage integration images; therefore, the jitter on images is the synthesized effect of the image at each integration stage. Since the motion induced by platform jitter varies over time and is influenced by the cumulative effects of motion at each stage of the multistage integration process of TDI CCD, the synthesized image differs from that produced by single-stage integration. The imaging time of each row is divided into N equal parts, where N corresponds to the TDI CCD integration stages.

According to the CDSM model [29], by taking the integration time of the first stage as the imaging time for multistage integration images, the image motion at row i under the influence of jitter can be formulated as

$$d_{\mathrm{TDI}}(i,m)=\frac{1}{mN}\left[\frac{1}{2}A\sin\left((i+N)\frac{2\pi\tau}{T}+\phi\right)+\sum_{k=1}^{mN-1}A\sin\left(\left(i+N+\frac{k}{m}\right)\frac{2\pi\tau}{T}+\phi\right)+\frac{1}{2}A\sin\left(i\cdot\frac{2\pi\tau}{T}+\phi\right)\right] \tag{1}$$

where τ is the single integration time of the TDI CCD and m is the subdivision numbers of the CDSM [29]. The value of m was set to 10 based on the previous work [13]. CDSM provides a denser sampling strategy to reduce associated errors. It divides the imaging time of a single row into $N \times m$ parts, resulting in a total of $N \times m + 1$ sampling points.

The values of the key parameters directly impact the imaging effect. A and f control the amplitude and frequency

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of the jitter, respectively, directly affecting the extent and frequency of pixel position changes; ϕ represents the phase, influencing the starting point of the jitter. By adjusting these parameters, the imaging effects of the TDI CCD under varying conditions can be simulated. Datasets with different levels of distortion are constructed by introducing dynamic platform jitter parameters (amplitude A, frequency f, phase ϕ), integration level N, and integration time τ. These three dynamic platform vibration parameters are randomly selected within a fixed range, adhering to the principle that higher frequencies correlate with smaller amplitudes and vice versa for lower frequencies.

In this article, we introduced jitter level L_J to describe different levels of jitter distortion by using the ratio of the amplitude attenuation factor to the amplitude. The reduction of jitter-induced errors over multiple integration stages in the TDI imaging system is expressed by the amplitude attenuation factor $\alpha = |\sin c(N\tau f)|$ [13]. Low-frequency jitter can be considered as the superposition of linear motion, leading to noticeable geometric distortion in images. Conversely, high-frequency jitter results in multiple jitter cycles within the imaging time, thereby intensifying the blur effect. With the main goal of correcting minor geometric distortions, the criterion for constructing the dataset is to simulate distorted images with lower levels of blurring. Since geometric distortion is primarily influenced by amplitude, we defined jitter levels based on the amplitude A and amplitude attenuation factor α. A dataset with a jitter level i can be defined as follows:

$$L_J^{(i)} = \frac{\alpha^{(i)}}{A^{(i)}}. \tag{2}$$

Through the above simulation method, a large number of distorted and undistorted image pairs can be generated for network training. Additionally, relevant information for each image (such as image name, jitter simulation parameters, etc.) is recorded in a table. These sample sets cover a range of geometric distortions, from subtle to severe. After pairing the simulated distorted images with their corresponding reference images, these image pairs are used for neural network training. By training on datasets with diverse levels of geometric distortion, the neural network is better equipped to correct images with different degrees of distortion that may occur in real-world situations.

B. Architecture of JitterCorrectionGAN

The relationship between images with and without jitter effect can be described as

$$\mathbf{I}(\mathbf{x}) = F(\mathbf{J}(\mathbf{x} + \mathbf{D}(\mathbf{x}))) \tag{3}$$

where $\mathbf{I}(\mathbf{x})$ represents the image without geometric distortion, $\mathbf{J}(\mathbf{x})$ is the observed image with geometric distortion, $\mathbf{D}(\mathbf{x})$ is the 2-D distortion field reflecting the geometric distortions caused by jitter, and $F(\mathbf{x})$ denotes the process of image detail restoration.

Fig. 3(a) illustrates the correction process. Considering that the original distorted image typically exhibits blurriness, the original image is first passed through the detail enhancement net to preprocess the distortion effects, thereby reducing the blurriness and clarifying the boundaries of the geometric regions in the image. The structure of the detail enhancement net is shown in Fig. 3(c). Then, the preprocessed distorted image is fed into the distortion net to generate a distortion field. Following this, a corrected image is obtained through bilinear interpolation. The corrected image is then passed through the detail enhancement net again to produce the final corrected image. This step restores details lost during resampling and preserves image quality. The structure of the distortion net is shown in Fig. 3(b). This network is designed to address imaging distortions caused by satellite attitude jitter along three axes. Such attitude jitter typically manifests as periodic motion or as a superposition of motions with different periods and frequencies, which can be decomposed into two vectors along the x- and y-axis on the image. The goal of the distortion net is to learn the distortion field of the distorted image, which contains displacement vectors for each pixel, indicating how each pixel should be shifted to achieve the desired transformation. After learning the distortion field, the original image is adjusted using bilinear interpolation to correct the distorted image. Although this process can significantly correct the geometric distortion of the image, it may lead to some information loss and image blurring. In order to address this, the detail enhancement net is used again to repair the details of the remote sensing image processed by the distortion net, thereby preserving the quality of the generated image.

The JCGAN model consists of a generator and a discriminator. The generator takes the distorted image as input and attempts to generate a corrected image to rectify the geometric distortion. The discriminator evaluates the differences between the generated and the reference images, outputting a score to assess the authenticity of the generated image. The generator and discriminator compete through adversarial training to enhance the generator's ability to correct the distortions. In Fig. 3(d), InstNorm refers to the instance normalization layer, and ReLU denotes the rectified linear unit. Both network architectures follow a similar pattern, specifically an encoder-decoder layout. The encoder of each network consists of three convolutional layers, each with a kernel size of 4, a stride of 2, and a padding of 1. Each convolutional layer is followed by an instance normalization layer and a ReLU activation layer. The output feature map dimensions produced by these layers are 32, 64, and 128, respectively. Correspondingly, the structure of the decoder is almost identical to that of the encoder. As shown in Fig. 3(f), in both networks, following the last transposed convolution layer, the output layer uses the Tanh activation function to limit the output values within the range of $[-1, 1]$, ensuring consistency with the value range of the input image. Additionally, as shown in Fig. 3(e), both network architectures integrate eight residual blocks between the encoder and decoder to facilitate learning of deeper features. Each residual block consists of two convolutional layers, each with a kernel size of 3, a stride of 1, and a padding of 1. Each convolutional layer is preceded by an instance normalization layer and followed by a ReLU activation layer. The detail enhancement net adopts a U-Net-like skip connection method to effectively fuse low-level and high-level features, thereby enabling the output image to retain more details.

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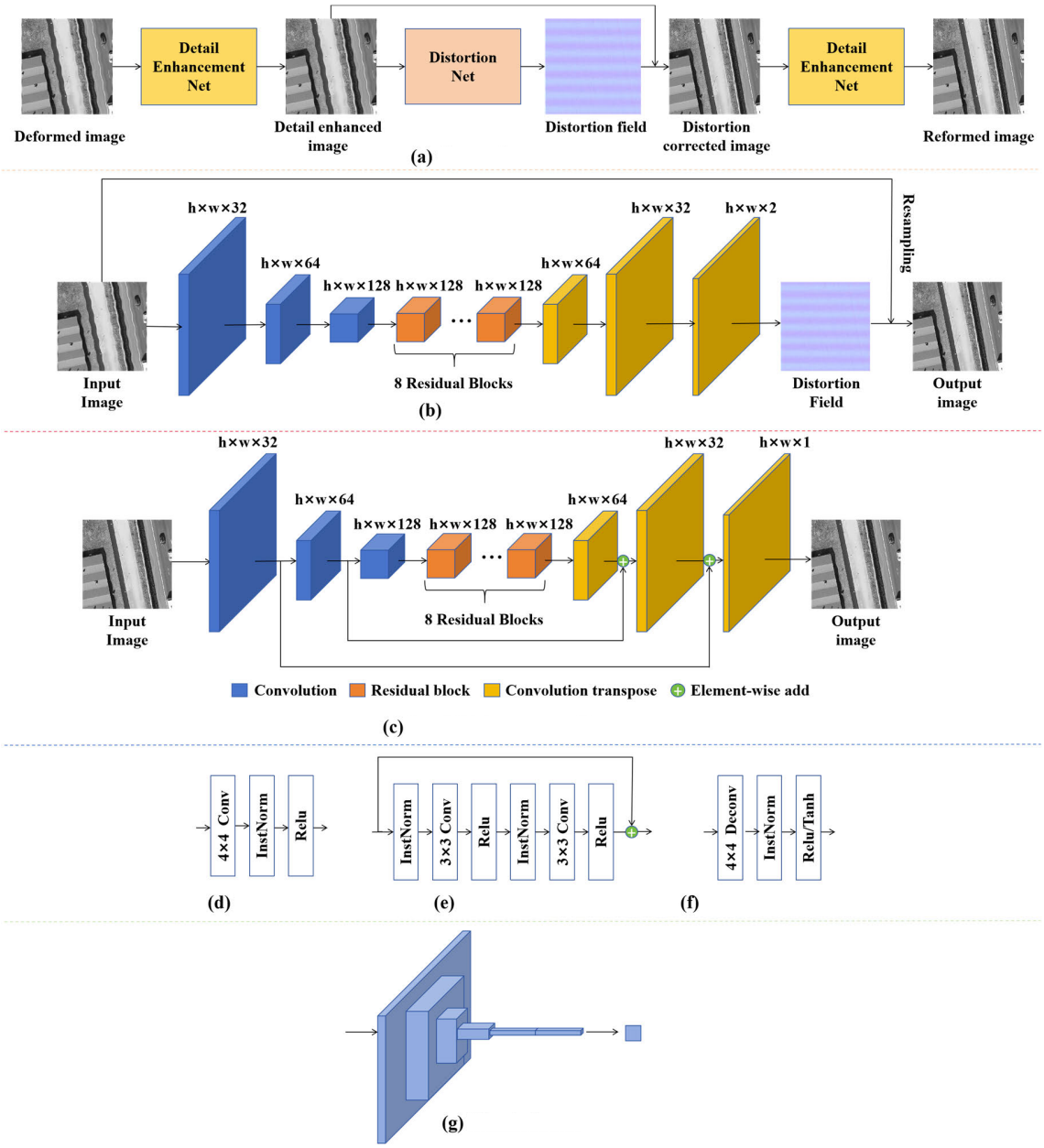


Fig. 3. Diagram of the JCGAN. (a) Correction process. (b) Distortion net. (c) Detail enhancement net. (d) Convolution. (e) Residual block. (f) Convolution transpose. (g) Discriminator.

The design of the discriminator is based on the discriminator structure proposed in DCGAN [43]. As shown in Fig. 3(g), it mainly consists of seven convolutional layers, each with a kernel size of 4, a stride of 2, and a padding of 1. The output feature map dimensions are 32, 64, 128, 256, 512, 512 and 1, respectively. Except for the last convolutional layer, each layer is followed by a batch normalization layer and a LeakyReLU activation layer. The last convolutional layer is connected to a Sigmoid activation layer, which outputs the probability of the image being real or fake.

C. Loss Function and Evaluation Metrics

The generator of network uses three types of loss functions: the content loss, the adversarial loss, and the perceptual loss. The $L1$ loss was selected as the content loss, which can be mathematically represented as follows:

$$L_{\text{con}} = \sum_{\mathbf{x}} |\mathbf{I}(\mathbf{x}) - G(\mathbf{J}(\mathbf{x}), \mathbf{x})| \tag{4}$$

where $\mathbf{I}(\mathbf{x})$ represents the ground truth, and $G(\mathbf{J}(\mathbf{x}), \mathbf{x})$ represents the image generated by the generator. Relying solely on the $L1$ loss as the training supervision criterion often results in blurry images and poor performance in reproducing high-frequency details. In order to address this issue, the discriminator loss is incorporated into the training process. This method significantly enhances the ability of the network to reproduce details, even simulating information loss resulting from significant distortions.

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The original GAN faces two major challenges. One is the vanishing gradient problem, which complicates training. The other one is the insufficient clarity of generated images. In order to address these issues, an improved algorithm known as the least squares GAN (LSGAN) [44] is adopted. This method is specifically designed to address the shortcomings of the original GAN. The adversarial loss function is given as follows:

$$L_{\text{adv}} = \frac{1}{2} E_{\mathbf{z}\sim p_{\mathbf{z}}(\mathbf{z})}\left[(D(G(\mathbf{z})) - 1)^2\right] \tag{5}$$

where $\mathbf{z}$ represents the noise vector.

Although LSGAN improves the representation of details in images, it may also introduce artifacts. In order to address this issue, the perceptual loss function is introduced. The definition of the perceptual loss function is shown as follows:

$$L_{\text{per}} = \sum_{\mathbf{x}} |\psi(\mathbf{I}(\mathbf{x}) - \psi(G(\mathbf{J}(\mathbf{x}), \mathbf{x}))| \tag{6}$$

where ψ represents the output of an intermediate feature layer selected from a pretrained neural network.

Specifically, the perceptual loss is calculated by inputting both the generated image and the reference image into a pretrained neural network, extracting their features at the same layer, and calculating the L1 or L2 distance between these two feature maps. This method focuses on computing the content differences in high-level features between the generated and reference images, thereby making the results more broadly applicable. The perceptual loss function has been successfully applied in fields such as image super-resolution [45] and deblurring [37], demonstrating its effectiveness in improving image quality.

Finally, the loss function of this method is a weighted combination of three losses, which can be formulated as

$$L = L_{\text{con}} + \lambda_{\text{adv}} L_{\text{adv}} + \lambda_{\text{per}} L_{\text{per}} \tag{7}$$

where λ_{adv} and λ_{per} are the weighting coefficients for the adversarial loss and perceptual loss, respectively.

Peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM) were selected as benchmark metrics to evaluate image restoration performances. PSNR is used to measure the noise level between the restored image and the reference image. Higher PSNR values indicate better restoration performance. SSIM is used to assess the structural similarity between the restored image and the reference image. SSIM values range from 0 to 1, the closer the value is to 1, the better the restoration effect.

D. Progressive Learning Strategy

The proposed correction method assumes that the perceptual capability of neural networks in processing remote sensing images with subtle geometric distortion needs to be gradually improved. If the network attempts to directly learn and correct images that exhibit low level of geometric distortion, it may struggle to effectively capture pixel-level deviations. We, therefore, propose a progressive learning strategy that gradually guides the model from correcting images with higher levels of geometric distortion to those with lower levels of geometric distortion.

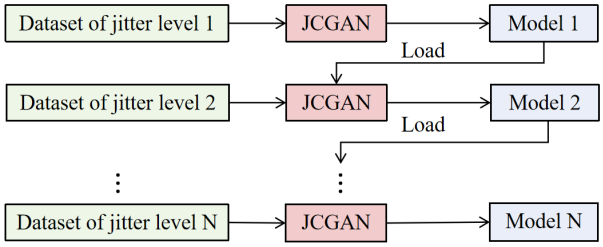


Fig. 4. Progressive training process.

This strategy divides the training process into multiple stages. As shown in Fig. 4, in the initial stage, the network focuses on learning to correct images with significant geometric distortion using a dataset primarily composed of highly distorted images. The purpose at this stage is to enable the network to effectively correct prominent distortions. Subsequently, the pretrained model from the previous stage is loaded. Further training is then performed using a dataset with a lower level of geometric distortion. This process enhances the model's ability to identify and correct less distorted remote sensing images. Following this, the pretrained model from this stage is used to continue training the network on a dataset with even lower distortion levels. By repeating this process, the network gradually refines its ability to detect and correct subtle geometric distortions caused by high-frequency jitter. This stage represents the most challenging part of the training strategy, as it requires the network to detect and correct subtle degradation phenomena that only occur under high-frequency jitter conditions.

Through this progressive training approach, learning from simple to complex, the detection and correction capabilities of network are gradually improved, especially for the intricate degradation phenomena caused by high-frequency jitter. Each pretrained model serves as a "template," guiding the next stage of training. In the subsequent stages, the network leverages the knowledge gained in the previous stage to refine its learning step by step. This strategy significantly enhances the network's ability to handle images with diverse levels of distortion, ensuring that it can effectively learn the distortion mechanism.

IV. Experiment and Discussion

First, we perform jitter simulation to reference images using the CDSM-based TDI CCD imaging model and generate datasets with diverse degrees of distortion by adjusting the simulation parameters. Subsequently, through a progressive learning strategy, the neural network gradually learns to correct subtle geometric distortions caused by high-frequency jitter. In order to validate the effectiveness of our proposed method, we conducted experiments on a desktop system running Linux, equipped with an Intel i9-13 900 CPU and an NVIDIA RTX A5000 GPU. The code was executed on this system configuration to perform the analysis and obtain results for evaluation. We compared our method with some typical deep-learning algorithms for image restoration. The images in

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Sections IV-B–IV-F have been scaled down for formatting purposes; please see the zoomed-in view for a better view.

A. Datasets Simulation With Different Jitter Levels for Learning and Testing

Considering that dataset selection plays a crucial role in the training process, we carefully collected and selected a large number of remote sensing images from datasets such as DOTAv1.5, DOTAv2.0 [46] and VEDAI [47]. After cropping and rigorous filtering, a total of 65 195 images were obtained. Each image has a size of 512 × 512 pixels. These images contain numerous regions with distinct geometric features, which effectively support geometric distortion correction tasks and provide a solid foundation for subsequent research.

All high-resolution Earth observation images captured by satellites in the orbit of 500∼800 km, however, suffered from different levels of degradation due to the platform motion, atmospheric conditions and other factors. This implies that a perfect reference remote sensing image does not exist for the network learning, which limits the performance on distortion correction of the model. Only regions with distinct geometric features, such as edges of buildings and road boundaries, can, furthermore, effectively reflect distortions. Areas like forests, farmland, and nonedge regions of roads and buildings are challenging for the network to capture and correct due to their lack of noticeable texture or features. It is difficult to learn sufficient information using remote sensing images lacking significant texture.

Taking these considerations into account, we assume that the specific content of the images is not the main factor in model training. On the contrary, the key lies in the motion changes exhibited by the distorted image relative to the reference image. As long as the distortion effects in natural images can simulate distortion characteristics similar to those in remote sensing images, they can be used as reliable dataset. By ensuring that natural images are distorted in a way that reflects the distortions found in remote sensing images, these natural images become suitable substitutes for remote sensing images in the training process. It is particularly useful when remote sensing data is scarce or expensive to obtain. Then, we integrated data from natural images to enhance the original dataset, supplementing the geometric information and texture features required for training. A large number of clear images with geometric features were collected and rigorously screened from datasets such as ImageNet [48], COCO [49], and Places2 [50]. This strategy was validated in subsequent experiments, showing that supplementing with these natural images improved the performance of the model and significantly enhanced the training outcomes. A total of 60 000 natural images were collected as the training set, 5000 remote sensing images as the validation set, and 1111 remote sensing images as the test set. The dataset we collected is named JitterSimalation Set.

According to (1), we simulated the image displacement curves caused by platform jitter at different jitter frequencies and integration stages. The imaging process of TDI CCD remote sensing images under satellite platform jitter was simulated using MATLAB. Simulation key parameters include the integration level N, time interval dt (set to 0.00036 s), initial time T_0, jitter amplitude A, frequency f, and phase ϕ. By setting these parameters, the code performs multiple jitter deviation calculations for each pixel position of the input image, simulating the positional changes during the actual imaging process. Specifically, the input image is initially converted to double precision, and a blank image is initialized to store the simulation results. For each pixel position, multiple jitter deviations are calculated based on the set parameters.

After parameter adjustment and analysis, we found that the amplitude attenuation factor of the real distorted images is approximately 0.9. We divided the dataset into five groups based on (2), using the parameters listed in Table I. First, the simulation based on 60 000 natural images was conducted by using $l_f^{(1)}$ parameters because it is easier for the network to detect and correct geometric distortion with the higher distortion level. In order to increase data diversity and enhance the generalization capability of the model, we expanded the dataset size by rotating the images counterclockwise by 90°, 180°, and 270°. After further experimentation, we found that increasing the training set size beyond 120 000 images did not lead to significant improvements in correction performance. As a result, we kept the training set size fixed at 120 000 images in subsequent experiments. According to the previous work [13], the image distortion caused by jitter will gradually decrease as the integration stages of the TDI CCD increase when the integration time and jitter frequency are fixed. We set the integration levels within the range of 12–24 levels for geometric distortion correction experiments.

B. Comparison Results With Other Restoration Methods

Given the lack of learning-based methods specifically designed for geometric distortion correction, as well as the absence of datasets dedicated to this task, we train and evaluate our proposed method with other learning-based image restoration methods on the JitterSimulation Set for comparison. Considering the necessity of models that support the loading of pretrained weights, we selected MIMO-UNet+ [31], Restormer [32], and Uformer [33] as competing methods. For a fair comparison, all competing methods were trained from scratch on our JitterSimulation Set, adhering to the original training protocols of each method unless otherwise specified. During testing, PSNR and SSIM were used as evaluation metrics with a full image size of 256 × 256. In order to compare the complexity between our model and other state-of-the-art methods, we calculate the multiply-accumulate operations (MACs) and parameters. As shown in Table II, the other methods did not achieve satisfactory correction results. We attribute the effectiveness of our method to the implementation of progressive training, which is guided by the distortion field. Specifically, we train the network on a dataset with a higher degree of distortion, enabling it to correct row-variant distortions. Subsequently, we replace the dataset with one that has a relatively lower distortion level and loads the pretrained model for the next training stage. Loading the pretrained model can be considered as a specialized form of "row-attention" mechanism, allowing the model to focus directly

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TABLE I
PARAMETERS OF DIFFERENT JITTER LEVELS FOR DATASET GENERATION

Jitter level	Range of $L_J^{(i)}$	Range of jitter period (pixel)	Range of jitter error amplitude of single integration(pixel)	Range of jitter phase(rad)	Range of amplitude attenuation factor α
1	[0.320,0.407]	[69, 79]	[2.0,3.0]	[-π, π]	[0.814,0.960]
2	[0.487,0.571]	[79, 92]	[1.5,2.0]	[-π, π]	[0.856,0.973]
3	[0.655,0.893]	[92, 111]	[1.0,1.5]	[-π, π]	[0.893,0.982]
4	[0.989,1.319]	[111,138]	[0.7,1.0]	[-π, π]	[0.923,0.989]
5	[1.417,1.906]	[138,185]	[0.5,0.7]	[-π, π]	[0.953,0.992]

TABLE II
QUANTITATIVE RESULTS OF COMPETING METHODS WITH PROGRESSIVE LEARNING STRATEGY ON SIMULATED TEST SETS

Dataset with different jitter level	MIMO-UNet + [31]		Restormer [32]		Uformer [33]		JCGAN(ours)	
	MACs	Parameters	MACs	Parameters	MACs	Parameters	MACs	Parameters
	154.58 G	16.11 M	141.24 G	26.13 M	86.39 G	50.88 M	44.21 G	23.14M
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
Dataset 1	18.229	0.414	17.937	0.424	20.663	0.631	32.945	0.968
Dataset 2	20.273	0.613	20.742	0.661	21.261	0.685	33.116	0.968
Dataset 3	21.270	0.667	21.856	0.696	21.954	0.713	35.408	0.976
Dataset 4	23.101	0.767	24.068	0.789	25.126	0.853	35.258	0.970
Dataset 5	27.934	0.921	28.225	0.926	26.125	0.887	37.238	0.986

on row-variant distortions in the subsequent training, thereby reducing training complexity and preventing the network from learning in the wrong direction. Our method requires fewer parameters and MACs, achieving a good balance between performance and efficiency. Fig. 5 presents the correction performance of our method compared to other methods on simulated datasets with the five jitter levels.

C. Real Data Experiment Results

We applied our correction method to correct images with diverse levels of jitter distortion captured by the high-resolution satellite with TDI cameras. Based on the severity of distortion, we classify the collected distorted images into four groups, labeled distorted image 1 to distorted image 4. Figs. 6–9 present the correction results of real distorted remote sensing images. Fig. 6 displays the distorted image 1, which exhibits considerable geometric distortion. Our method demonstrates an effective correction in this case. Fig. 7 shows the distorted image 2, and Fig. 8 shows the distorted image 3.

Although the geometric distortion in these images is less pronounced than that in Fig. 6, they are characterized by a higher level of blur. Nonetheless, our method still yields effective correction results. Fig. 9 presents the distorted image 4, which exhibits subtle geometric distortion but a significantly high level of blur. In spite of this, our approach remains capable of effectively correcting the distortion. It can be observed that our method achieves notable correction performance, whereas other methods show limited effectiveness and, in some cases, even exacerbate the distortion. Our approach not only effectively corrects geometric distortions but also preserves image quality. Additionally, our method can correct geometric distortion without relying on additional distortion information.

D. Comparison Results With and Without Progressive Learning

We use natural images as the training set and remote sensing images as the validation and test sets to evaluate the correction performance of the model on remote sensing images. The reason for this classification is that natural images generally exhibit higher quality, with more abundant geometric features and textures, compared to remote sensing images. Training the network on natural images is more beneficial for learning. Remote sensing images are used as the validation and test sets because the ultimate goal of our experiment is to perform correction on remote sensing images. Remote sensing images are, therefore, employed as the validation set to verify whether training with natural images can effectively correct distortions in remote sensing images. Remote sensing images are also used as the test set to evaluate the model and select the one yielding the best correction results. We will use the model that achieved the best correction performance on the test set to correct the real distorted remote sensing images. Additionally, we also tried mixing remote sensing and natural images in specific ratios to create a training set. The evaluation results, however, showed that using a model trained on the mixed dataset performed worse than the model trained solely on the natural image dataset.

Fig. 10 shows the visualization of the progressive training process. The results are shown in Fig. 11. When the

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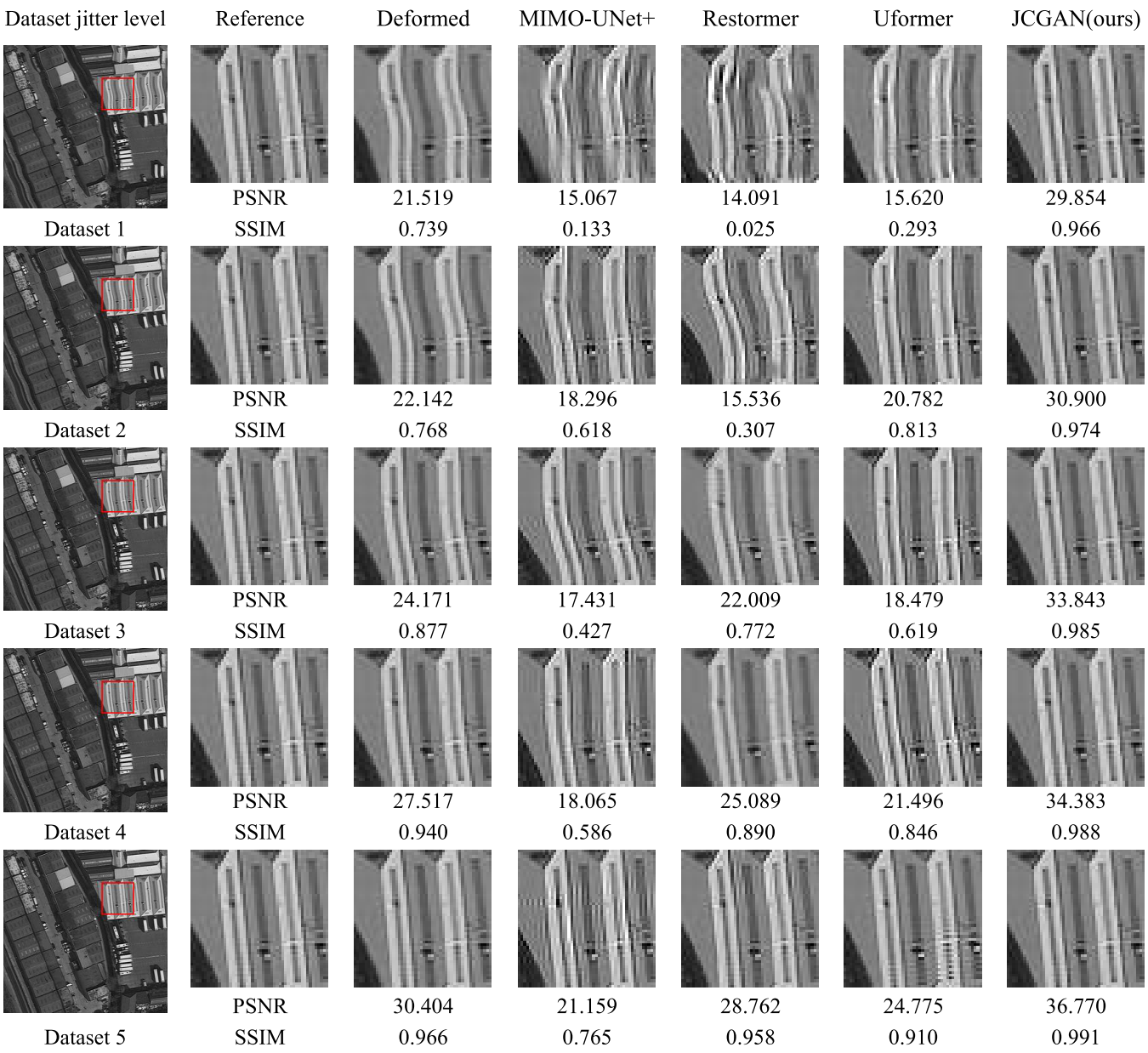


Fig. 5. Correction performance of competing methods with progressive learning strategy on simulated test sets.

progressive learning strategy is not applied, the PSNR and SSIM values of the reformed images are even lower than those of the deformed images, indicating a failure in model training. In contrast, our method achieves significantly better correction results when the progressive learning strategy is employed. We also trained the model on the five datasets with different jitter levels simultaneously and evaluated the results from this training. The model was evaluated using five test sets corresponding to the five jitter levels from Table I to assess whether it can correct images with diverse levels of distortion or to determine which distortion level it can effectively correct. The evaluation results are shown in Fig. 12. It can be observed that neither our method NOR the comparison methods could effectively correct the distortions.

Our progressive learning method fully considers the learning characteristics of neural networks. Just like human learning, it starts from the basics and progresses step by step. We first use dataset 1 to train the model, as it exhibits a higher level of distortion, which facilitates the network in performing corrections and developing an awareness of row-oriented correction. From the distortion field shown in Fig. 10(b), it is evident that during the training with dataset 1, the network has already developed the ability to correct row-variant distortions. By loading a pretrained model, we not only initialize the model parameters but also ensure that the model begins with an awareness of correcting row-variant distortions in the subsequent stages. The correction range of the model is continuously refined during training, enabling it to handle smaller geometric distortions. By repeating the above training process, we gradually enhance the ability of the model to correct subtle geometric distortions, ultimately enabling it to correct real distorted remote sensing images in

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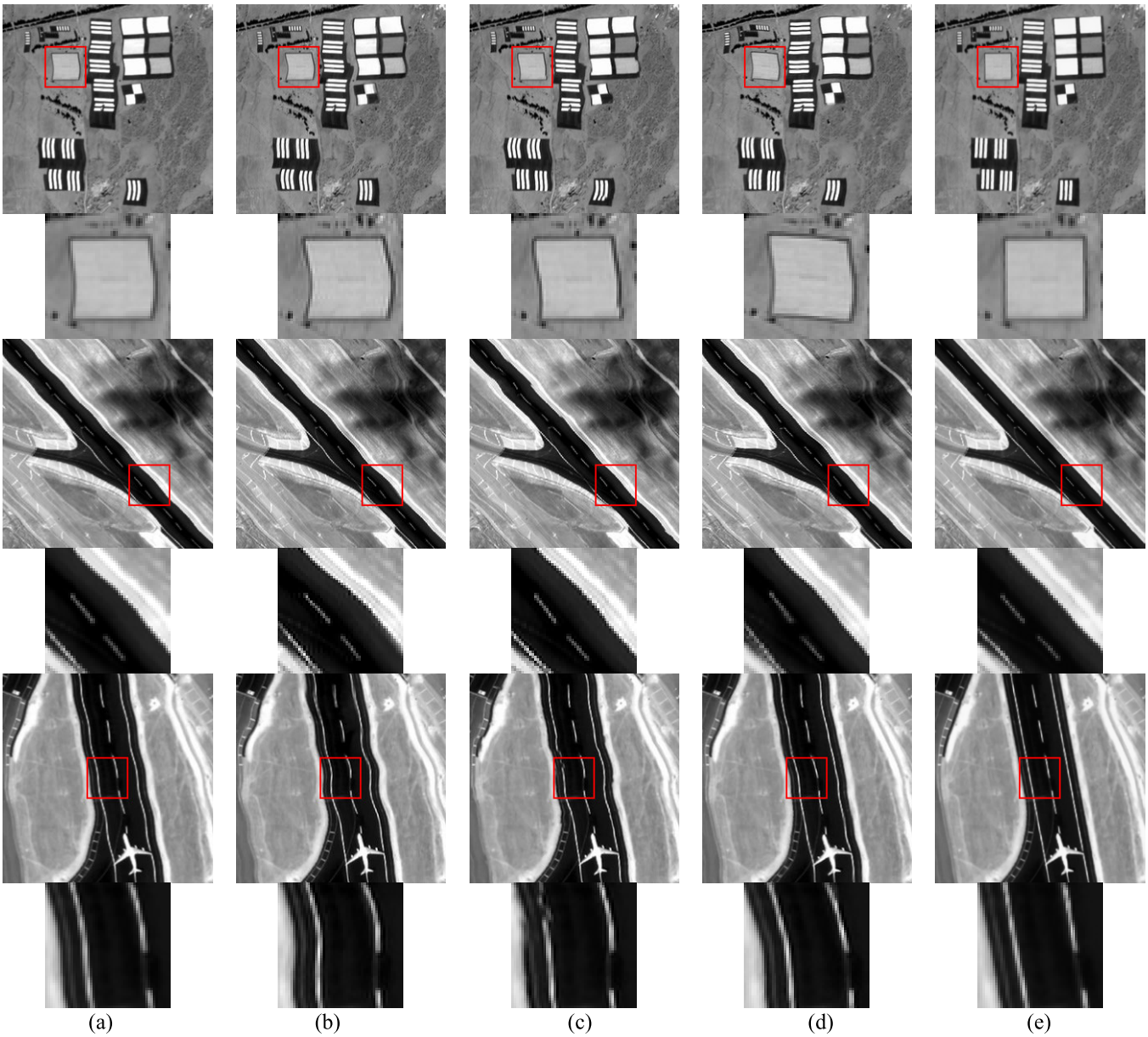


Fig. 6. Correction results on the distorted image 1. The results were achieved by Model 2, which was trained based on the pretrained Model 1. (a) Origin image. (b) MIMO-UNet+. (c) Restormer. (d) Uformer. (e) Our method.

practical applications. As shown in Fig. 11, when the model is trained directly on dataset 2 without loading a pretrained model, the PSNR and SSIM values of the corrected images are lower than those of the original distorted images.

E. Ablation Study

In order to demonstrate the effectiveness of the two networks used for correction, we conduct an ablation study on the network structure using the simulated dataset. Specifically, we sequentially used datasets 1–5 with a progressive training strategy to conduct ablation experiments. The evaluation results are shown in Table III; we can conclude that the complete network achieves better correction performance. In order to more clearly show the results of the ablation study, we displayed the correction results obtained during this experiment. Fig. 13 shows the results of the ablation study on the simulated test set from dataset 1. We selected the test set of dataset 1 for evaluation because the images exhibit a higher degree of distortion, which facilitates our analysis of the experimental results. The first column presents the simulated distorted images, which exhibit considerable geometric distortions. The second column presents the correction results after removing the distortion net from the model, showing that geometric distortions are still clearly visible. The third column shows the correction results without the detail enhancement net, where image clarity remains insufficient. The fourth column presents the results from the complete model, which includes both the distortion net and the detail enhancement net. the fifth column displays the ground truth. Through ablation experiments, we conclude that the distortion net plays a critical role in the correction process, particularly

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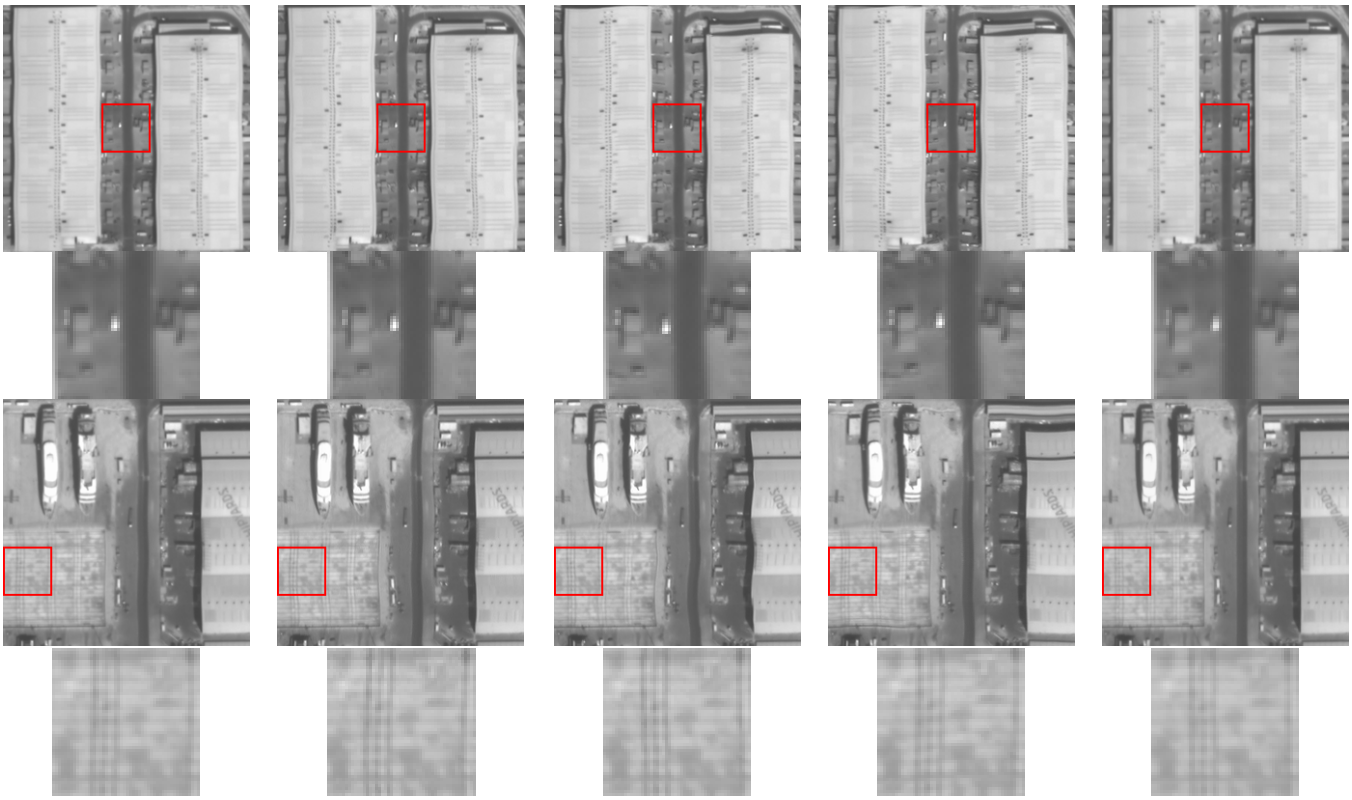


Fig. 7. Correction results on the distorted image 2. The results were achieved by Model 3, which was trained based on the pretrained Model 2. (a) Origin image. (b) MIMO-UNet+. (c) Restormer. (d) Uformer. (e) Our method.

in removing geometric distortions. Meanwhile, the detail enhancement net effectively enhances image details, thereby preserving image quality. The combined effect of these two networks, leveraging their complementary functions, greatly enhances the performance of the correction task.

F. Discussion

1) Performance Comparison Using Additional Metric: In our study, we primarily use the traditional PSNR and SSIM as image quality assessment (IQA) metrics for paired datasets. For real distorted remote sensing images, however, there are no corresponding reference images without distortion, which prevents the use of PSNR and SSIM for quality assessment. We therefore, introduce multiple pseudo reference image (BMPRI) measure [51] to further evaluate the achieved restoration performance. The BMPRI is a no-reference IQA (NR-IQA) method that generates multiple pseudo-reference images by aggravating distortions in the distorted image. It enables image quality estimation without relying on reference images and can sensitively reflect content sharpness and noisiness. The higher the BMPRI value, the better the restored image quality. The evaluation results are shown in Table IV. It is noteworthy that JCGAN outperforms the competing methods, further demonstrating the effectiveness of our method.

2) Parameters Study: In the experiments, we used test set from dataset 1 to perform fine-tuning of the loss function weights. It was found that setting the values of λ_{per} and λ_{adv} to 0.5 and 0.2 respectively can yield the best correction effect. The results are shown in Tables V and VI.

Additionally, we selected the conv4_3 feature layer in VGG16 as ψ in the perceptual loss.

3) Comparison With JARNet: JARNet [40], [41] is the most recent and highly relevant works in this field. The training process of JARNet relies on the jitter information from the simulation to generate a jitter map, which is then used to construct the optical flow for geometric distortion correction. In contrast, our method is fundamentally different, and it relies solely on paired distorted and undistorted images and employs a distortion net to directly generate the distortion field for correction. Because of the differences in experimental design between the two approaches, a direct and fair comparison under the same conditions is not feasible. It is, however, worth noting that Chen et al. [40] and Zhang et al. [41] validated their method using real distorted images from the Yaogan-26 and achieved favorable results. Considering this, we compare the correction performance on the images from Yaogan-26 between our method and JARNet. As shown in Fig. 14, our method achieves more notable geometric correction, while the results from JARNet still exhibit slight residual distortions. We further evaluated the comparison results using the BMPRI measure [51]. Across the three images, our method achieved a higher average BMPRI value of 25.054, compared to 18.308 for JARNet. Nonetheless, the images restored by JARNet exhibit higher visual clarity. Although JCGAN achieves better performance in geometric distortion correction, its capability in enhancing image quality remains insufficient.

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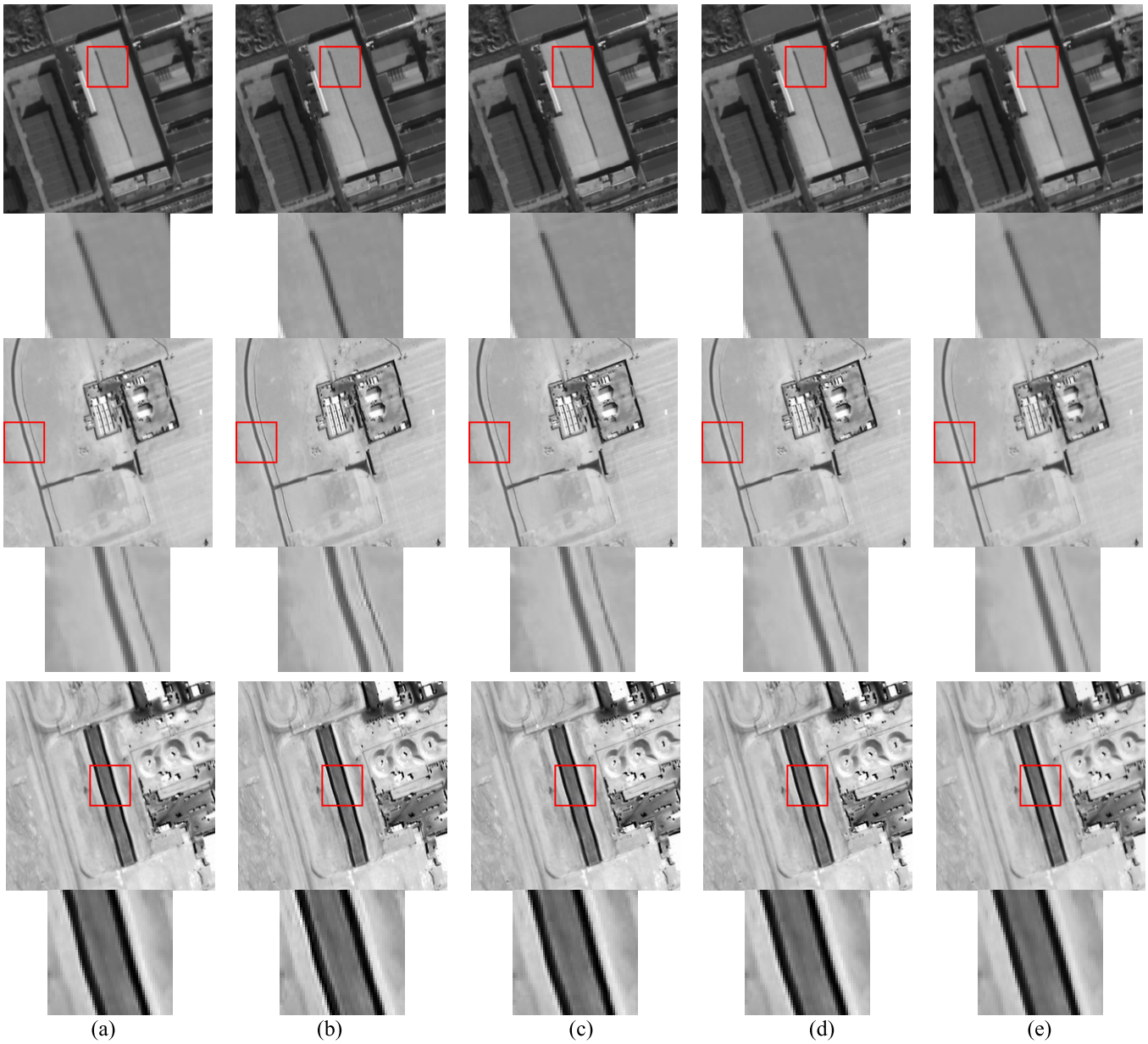


Fig. 8. Correction results on the distorted image 3. The results were achieved by Model 4, which was trained based on the pretrained Model 3. (a) Origin image. (b) MIMO-UNet+. (c) Restormer. (d) Uformer. (e) Our method.

TABLE III
QUANTITATIVE RESULTS FOR ABLATIVE STUDY

Dataset with different jitter level	Without Distortion Net		Without Detail Enhancement Net		Full net	
	MACs	Parameters	MACs	Parameters	MACs	Parameters
	22.28 G	11.92 M	22.18G	11.23 M	44.21G	23.14M
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
dataset_1	29.579	0.915	32.602	0.957	32.945	0.968
dataset_2	32.217	0.955	33.219	0.964	33.116	0.968
dataset_3	33.738	0.966	33.147	0.962	35.408	0.976
dataset_4	35.184	0.978	34.440	0.972	35.258	0.970
dataset_5	36.815	0.981	36.278	0.982	37.238	0.986

Li et al. [39], Chen et al. [40], and Zhang et al. [41] also proposed data synthesis methods that are similar to the simulation method proposed in our paper. But there are some differences between these data synthesis approaches. First,

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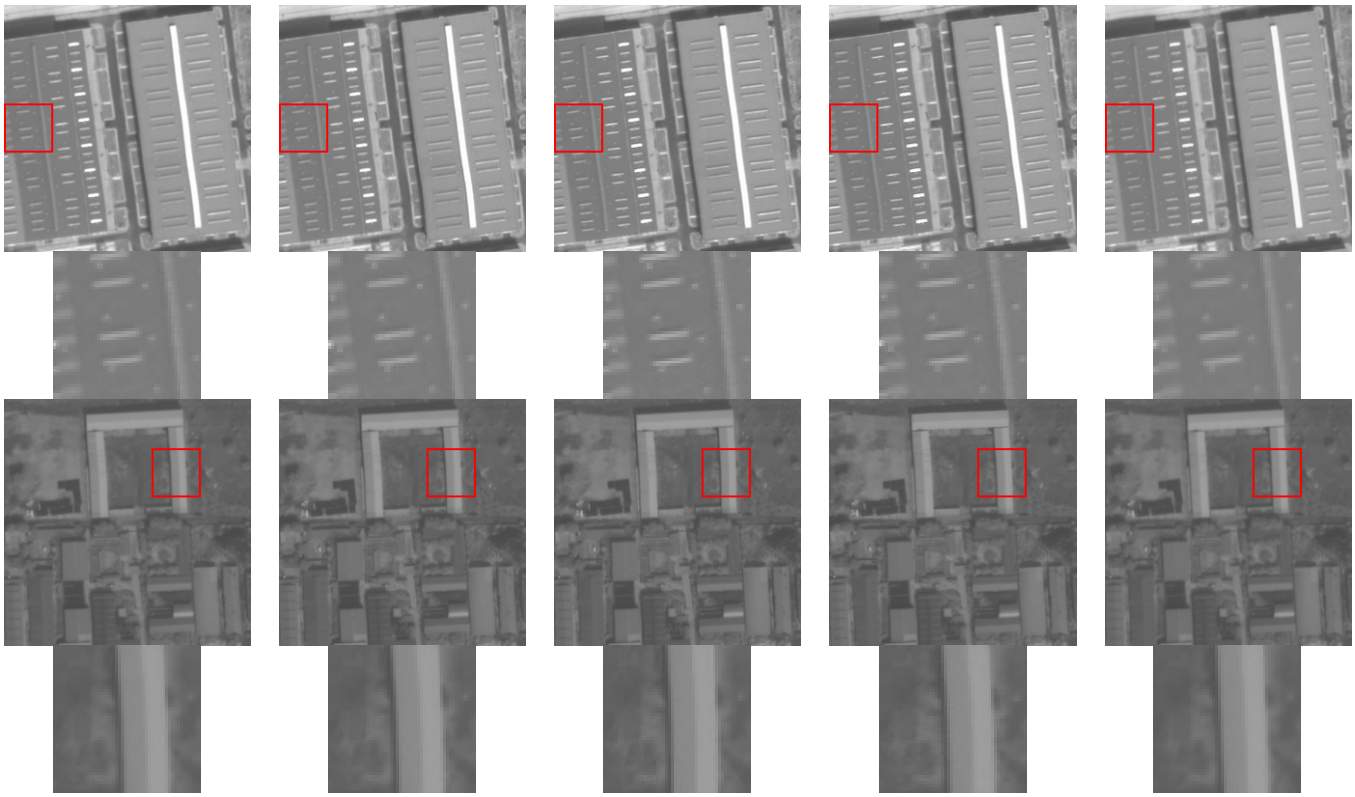


Fig. 9. Correction results on the distorted image 4. The results were achieved by Model 5, which was trained based on the pretrained Model 4. (a) Origin image. (b) MIMO-UNet+. (c) Restormer. (d) Uformer. (e) Our method.

TABLE IV
BMPRI EVALUATION RESULTS FOR REAL DISTORTED IMAGES

Image group number	MIMO-UNet+ [31]	Restormer [32]	Uformer [33]	JCGAN (ours)
1	13.173	15.795	17.520	29.467
2	19.416	20.401	19.527	27.027
3	12.356	16.425	18.909	26.402
4	16.012	13.685	13.129	23.577

TABLE V
PSNR EVALUATION RESULTS FOR WEIGHTING COEFFICIENTS

λ_{adv} \ λ_{per}	0.2	0.5	0.7	1
0.2	28.032	**32.945**	31.105	32.070
0.5	30.082	30.130	26.516	31.181
0.7	28.581	27.905	30.158	30.341
1	26.732	25.711	27.805	28.408

TABLE VI
SSIM EVALUATION RESULTS FOR WEIGHTING COEFFICIENTS

λ_{adv} \ λ_{per}	0.2	0.5	0.7	1
0.2	0.945	**0.968**	0.955	0.955
0.5	0.938	0.941	0.919	0.947
0.7	0.927	0.931	0.938	0.946
1	0.901	0.909	0.922	0.932

regarding the data source, our synthetic dataset comprises not only remote sensing images from datasets such as DOTA-v1.0 but also natural images from datasets such as ImageNet. Second, the dataset we constructed is a categorized dataset based on the jitter level. Third, although our method and the approaches in [39], [40], and [41] are based on the CDSM framework, our synthesis process focuses on platform-induced jitter and does not consider other degradation factors such as noise and photoelectric charge transfer time deviation. The jitter frequency, amplitude, phase, and TDI stages in our simulation method are, moreover, variable, with their values randomly selected within fixed ranges. As satellite resolution continues to improve, the effects of jitter induced by factors are becoming more significant; these factors will be incorporated into future versions of our simulation method.

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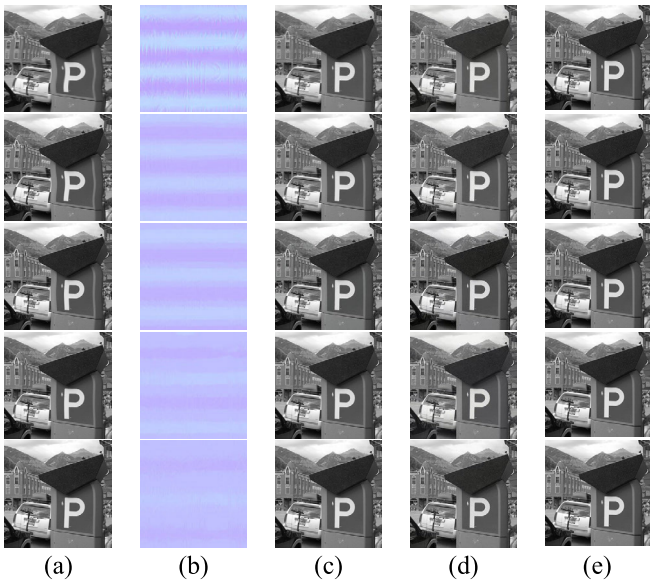


Fig. 10. Visualization of the progressive training process from jitter level 1 to 5. (a) Input distorted images. (b) Distortion field. (c) Corrected images obtained by applying the distortion field. (d) Results after applying detail enhancement net. (e) Ground truth.

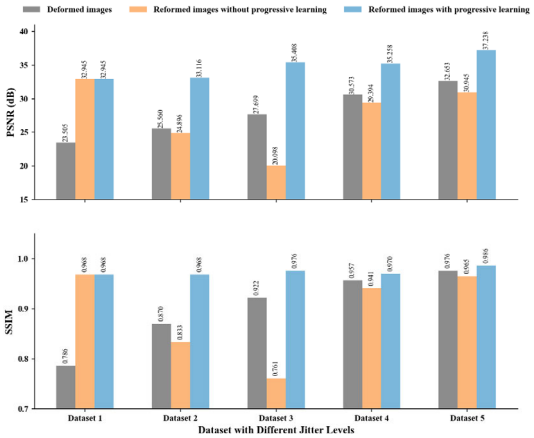


Fig. 11. Evaluation with and without progressive learning.

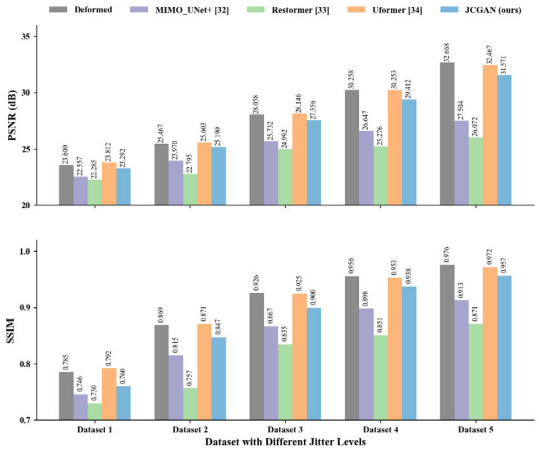


Fig. 12. Evaluation of training on five datasets simultaneously.

4) Limitations of JCGAN: The effectiveness of JCGAN in correcting geometric distortion relies on its ability to accurately predict pixel displacements. Consequently, its performance is less robust in regions with indistinct geometric features, such as water surfaces, forests, farmlands, roads, and nonedge areas of buildings. The reason is that the lack of distinct geometric structures in these regions makes it difficult to estimate pixel displacements accurately.

As shown in Fig. 15, two areas were selected for further analysis. In the second image, most of the area is covered

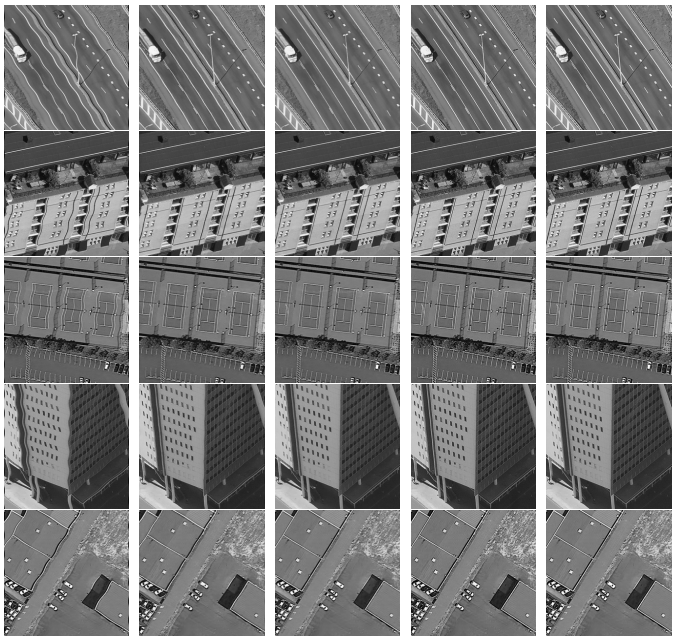


Fig. 13. Results for ablative study on simulated test set. (a) Simulated distorted image. (b) Correction result after removing the distortion net from the model. (c) Correction result after removing the detail enhancement net from the model. (d) Correction result of our complete network. (e) Ground truth images.

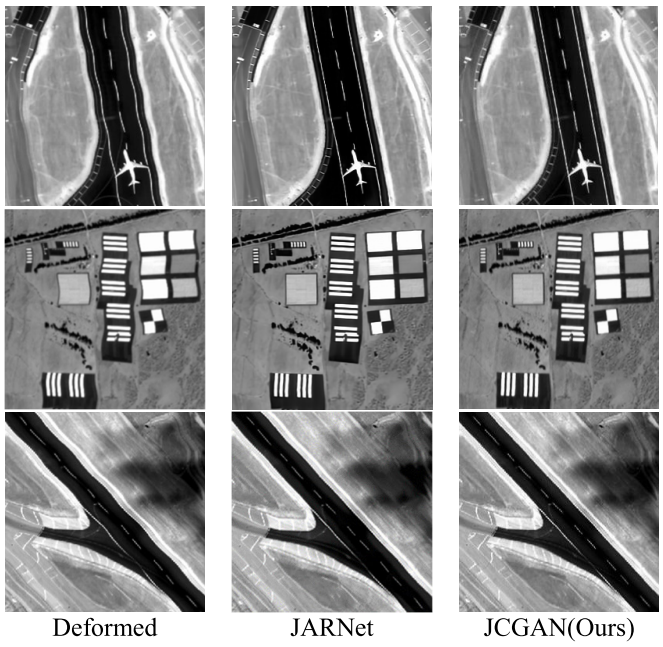


Fig. 14. Comparison with JARNet correction results of real-world distorted images from Yaogan-26.

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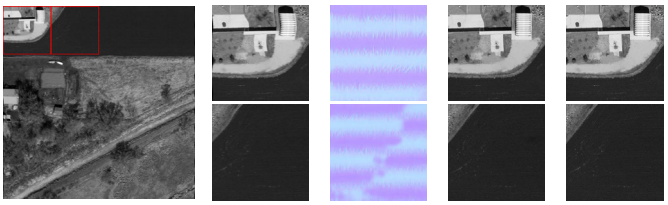


Fig. 15. Limitations of JCGAN in distortion field estimation. (a) Simulated distorted images. (b) Distortion field. (c) Reformed images (d) Ground truth images.

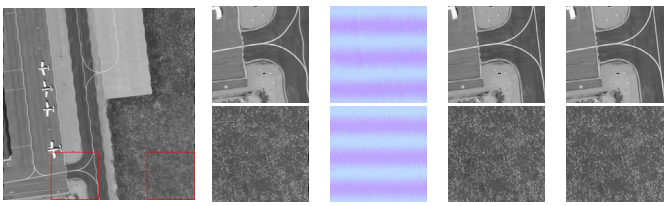


Fig. 16. Limitations of JCGAN in image quality restoration. (a) Simulated distorted images. (b) Distortion field. (c) Reformed images (d) Ground truth images.

by water, with only a small portion of land in the upper-left corner. The corresponding distortion field indicates that the distortion net failed to capture the correct deformation, leading to inadequate geometric distortion corrections.

As shown in Fig. 16, two regions were selected for analysis. In the second image, the entire area consists of grassland. Although the distortion net is able to generate a relatively accurate distortion field, the corrected image is noticeably blurred. Quantitative evaluation shows that the PSNR and SSIM of the first corrected image are 30.485 and 0.927, respectively, while those of the second corrected image are only 27.292 and 0.825. Additionally, due to the high degree of distortion in the images, the restoration results were suboptimal and even generate false textures.

Although JCGAN shows impressive performance in correcting geometric distortion, it currently does not enhance image quality but only preserve the original quality. Improving the quality of the restored image will be the priority of our future work.

V. Conclusion

In this article, we proposed a progressive training strategy to address the challenge of correcting subtle geometric distortions in satellite images caused by platform jitter. We attribute the effectiveness of our proposed method in correcting minor geometric distortions to the following reasons.

1) Recognizing the critical role of dataset selection in achieving optimal training results, we have specifically constructed a high-quality dataset for the geometric distortion correction task. We found that the image content is not as important as the motion changes exhibited between distorted and reference images. We, therefore, innovatively use high-quality natural images as our training dataset, thus avoiding the inherent flaws of remote sensing images.
2) The proposed method introduces a distortion net for geometric distortion correction and a detail enhancement net for restoring image details. From the experimental results, it can be seen that this network structure design can both correct image distortion and preserve image details.
3) We proposed a progressive training strategy for row-variant distortion correction in TDI CCD images. This learning approach not only facilitates weight initialization for each subsequent training session, but also enables the network to develop the ability to correct row-variant distortions from the early stages of training. Through continuous training and progressive refinement, this approach effectively corrects subtle distortions and balances distortion correction with detail preservation, which is crucial for high-resolution satellite imagery.

JCGAN shows competitive performance in correcting geometric distortions. It is noted that each model generated by JCGAN is limited to correcting distorted images that align or are similar to the jitter level in its training set. This implies that a preliminary evaluation of the jitter level of the distorted image should be conducted before correction. Addressing these limitations will be a key direction of our future efforts.

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